

Artificial Intelligence Fundamentals

Sample knowledge-base document for a Retrieval-Augmented Generation project.

What is Artificial Intelligence?

Artificial Intelligence (AI) is the field of building computer systems that perform tasks associated with human intelligence. These tasks include reasoning, learning from data, understanding language, recognizing patterns, planning, and making decisions.

Major Areas of AI

Common areas include machine learning, natural language processing, computer vision, robotics, knowledge representation, search, planning, and expert systems. Modern AI systems often combine several of these areas.

Machine Learning

Machine learning is a branch of AI in which models learn patterns from data instead of relying only on manually written rules. Supervised learning uses labeled examples, unsupervised learning discovers structure in unlabeled data, and reinforcement learning learns through interaction and rewards.

Training and Inference

Training is the process of adjusting model parameters using data. Inference is the process of using a trained model to make predictions or generate outputs on new inputs. Training is usually more computationally expensive than inference.

Evaluation

AI models are evaluated using metrics that match the task. Classification tasks may use accuracy, precision, recall, and F1 score. Regression tasks may use mean absolute error or root mean squared error. Good evaluation also requires checking performance on data not used for training.

Risks and Limitations

AI systems can make mistakes, inherit bias from data, fail on unfamiliar inputs, or produce overconfident answers. Human oversight, careful testing, privacy protection, security controls, and monitoring are important in real deployments.